MILITARY SPECIFICATION

CASE, CARTRIDGE, 20MM, XM204E1 (STEEL)

1. SCOPE

1.1 This specification covers the Steel Cartridge Case for use in the assembly of 20MM, Cartridges.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-651	- Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

D10542497	- Case, Cartridge, 20MM, XM204E1
C10542736	- Classification of Defects for Case, Cartridge, 20MM, XM204E1
D10542655	- Cartridge, 20MM, Target Practice, Tracer, M206E2
F11731525	- Carton, Interplant Shipment, 20MM, XM204E1, Cartridge Cases, Empty, Unprimed (100)
D11731526	- Shipping Container for Cartridge Cases, Unprimed, 20MM, (6000)
IEL-10542497	- Index of Inspection Equipment Lists for: Case, Cartridge, 20MM, XM204E1

FSC 1305

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2.1 (Cont'd)

PUBLICATIONS

U.S. Army Materiel Command

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117-64 - Method of Test for Salt Spray (Fog) Testing
ASTM E18-67 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM A317-54 - Macroetch Testing and Inspection of Steel

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with Drawing (Dwg) D10542497 referenced specifications, and other requirements specified herein.

3.2 Manufacturing process.- The cartridge case shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (See 6.2)

3.3 Hardness.- The cartridge case shall comply with Rockwell hardness pattern specified by Dwg. D10542497.

3.4 Cold shuts.- The cartridge case shall be free of cold shuts.

3.5 Protective coating.- Protective coating as specified on the applicable drawings and referenced specifications shall apply to all surfaces of the cartridge case regardless of contour configuration.

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3.5.1 Corrosion resistance.- The cartridge case shall show no visible signs of white corrosion products on exterior surfaces including accidental scratches through the surface film to the zinc plate after exposure to a 5 percent solution of atomized salt spray for 48 hours.

3.6 Maximum profile.- The cartridge case shall freely enter the maximum profile gage in such a manner as to permit the slide to close without interference.

3.7 Function and casualty.- The cartridge case, loaded and assembled in accordance with the requirements of Dwg. #D10542655, shall function without casualty at ambient temperature.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.- The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

3.8.1.1 Flow of metal.- The flow of metal in the section of the cartridge case joining the head and sidewall shall follow the direction of the contour of the interior curved surface without presence of folds, cold shuts or roughness.

3.8.2 Burr.- No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to the personnel handling the cartridge case.

3.8.3 Foreign matter.- No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms.

4.2 First article sample. (See 6.3)

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4.2.1 Initial production sample.- At the beginning of regular production, a sample applicable to the approved process shall be submitted in accordance with contract requirements and shall consist of:

a. Inspection and non-firing test(s)

100 cartridge cases

b. Firing test(s)

4000 cartridge cases

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. (See 6.2)

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract. Identification shall be in accordance with MIL-STD-1168.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions.-

4.3.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.- A lot shall consist of cartridge cases produced by one manufacture in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.3.1.2 Lot identification.- Each cartridge case and each packed cartridge case lot shall be identified in accordance with applicable drawings and MIL-STD-1168.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each cartridge case, found to be defective, shall be rejected.

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4.3.2.1 Sampling and acceptance criteria.- Except as otherwise indicated, sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the Acceptable Quality Level (AQL) assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, when sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects. (see 6.4)

4.3.2.2 AQL's.- The following AQL's are assigned:

a. Individual defect characteristic basis

Major defects	0.25%
Minor defects	0.40%

b. Defect characteristic, class basis

Major defects	1.50%
Minor defects	2.50%

c. Individual defect characteristic associated with the class basis

Major defects	0.40%

4.3.3 Classification of defects[1/] - The classification of defects shall be as follows:

4.3.3.1 Case, Cartridge (See Dwg D10542497)[1/]

Categories and Defects	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Total length, max.	Gage
102. Thickness, head rim, min.	Gage

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

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4.3.3.1 (Cont'd)

Categories and Defects.	Method of Inspection.
MAJOR	
MAJOR	
103. Width, extractor groove, min.	Gage
104. Diameter of head, min.	Gage
105. Diameter, extractor groove, max.	Gage
106. Diameter, primer pocket	Gage
107. Concentricity, primer pocket with datum surface	Gage
108. Depth, primer pocket	Gage
109. Profile and alignment, max.	Gage
110. Thickness, head rim	Gage
111. Diameter, top of body taper, min.	Gage
112. Diameter, bottom of body taper, min.	Gage
113. Wall thickness, .200 from inside base	Gage
114. Wall thickness, 1.00 from inside base	Gage
115. Wall thickness, 3.890 from inside base	Gage
116. Length to .961 diameter basic	Gage
117. Wall thickness at neck area	Gage
118. Inside diameter mouth, min.	Gage
119. Metal defective	Visual
120. Corrosion	Visual
121. Oil or grease	Visual
122. Head configuration	Visual
123. Chamfer, head rim missing	Visual
124. Flash hole missing or obstructed	Visual
125. Draw scratch	Visual
126. Protective finish missing or incomplete	Visual
127. Split or perforated case	Visual
MINOR	
201. Total length, min.	Gage
202. Diameter, head max.	Gage
203. Diameter, extractor groove, min.	Gage
204. Web thickness, min.	Gage
205. Diameter of flash hole	Gage
206. Width extractor groove, max.	Gage
207. Inside diameter of mouth, max.	Gage
208. Weight	Scale
209. Scratch (other than draw scratch)	Visual
210. Fold, wrinkle or buckle	Visual
211. Radius at mouth of primer pocket missing or not tangent to pocket side wall	Visual
212. Marking incorrect, incomplete or unidentifiable	Visual
213. Dent	Visual
214. Foreign matter other than oil, grease, or corrosion	Visual

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4.3.4 Tests.- The tests listed in Table I shall be performed on each cartridge case lot in accordance with the test methods prescribed in 4.4. Sample size and acceptance criteria for each test shall be as specified. Only cartridge cases having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cases shall be thoroughly mixed before being divided into samples for the various tests.

TABLE I

Test		Sample Size	Rqmt Para.
Hardness	1/	13	3.3
Cold shut	2/	13	3.4
Corrosion resistance	3/	13	3.5.1
Function and casualty	4/	400	3.7

1/ Failure of two or more of the sample cartridge cases to comply with the average hardness requirements shall cause rejection of the lot. If one of the sample cases fail to comply, a second sample of 13 cartridge cases shall be randomly selected from the lot and tested. The lot shall be rejected if a total of two or more cartridge cases of the combined first and second samples fail to comply with the applicable hardness requirement.

2/ The lot shall be rejected if one or more units of the sample shows a cold shut extending beyond the area identified by tangents AB and CD of Figure 1.

3/ Failure of two or more units of the sample to comply with the corrosion resistance requirement shall be cause for rejection of the lot. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

4/ The lot will be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable rejection ("Rej") number. If malfunctions or casualties occur in excess of the applicable acceptance ("Acc") number, but less than the "Rej" number, a second sample of double the number of the first sample. The lot will be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

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4.3.4.1 Firing defects.- Firing defects and associated criteria shall be as specified in Table II. For defect definitions, see AMCR 715-505, Volume 8.

TABLE II

Firing Casualties	Ambient Acc.	Ambient Rej.
Failure to chamber 1/	0	1
Failure to extract 2/	0	1
Blown primer 3/	0	1
Split 4/		
I or S	6	17
J	2	7
K, L or M	0	1
Rupture, partial		
S, J or K	2	7
L	0	1
Rupture, circumferential separation	0	1
Detached metal 5/	0	1

1/ The case lot will not be penalized if malfunction cannot be shown to have resulted from a defective case.

2/ A fired sample cartridge case remaining in the chamber after the normal extraction cycle will be classed defective, unless it can be shown that the malfunction was caused by a defective weapon.

3/ The sample cartridge case shall be classed defective only if it is evident by visual inspection that both the primer pocket and case head are enlarged and deformed.

4/ Refer to Dwg. C10542736 and AMCR 715-505, Vol. 8 for classifying splits and ruptures resulting from the function and casualty test. If a split or partial rupture extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved shall be used.

5/ The case lot will not be penalized if malfunction cannot be shown to have resulted from a defective case. There must be evidence of detached metal from fired case exterior. The lot will not be penalized for the detachment of metal shavings from the projectile crimped area of the case interior.

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4.3.4.2 Unlisted firing defects.- The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification, occurs in any firing test.

4.3.5 Packaging and marking inspection.- Inspection for packing and marking shall be in accordance with MIL-STD-1169.

4.3.6 Inspection equipment.- Index of Inspection Equipment List No. IEL 10542497 identifies the applicable Inspection Equipment List required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding format).

4.4 Test methods and procedures.

4.4.1 Hardness.- The cases of the test sample(s) will be sectioned as shown on Dwg. D10542497. The head sections shall have approximately parallel surfaces finished to 63 microinches or finer, care being taken not to change the hardness in obtaining this finish. Four determinations each will be made at the head, body and mouth positions for each sample case, two on each half case section. The method of test shall be as prescribed in American Society for Testing and Material (ASTM) E18.

4.4.2 Cold shut.- The method of test shall comply with the requirements of American Society for Testing and Material (ASTM) A317. The specimens for test shall be the head sections from the sample prepared for determination of the hardness requirements of 4.4.1. Macroscopic examination of the etched test specimens for cold shuts within the area identified by Figure 1 shall be made under 75-100 magnifications.

4.4.3 Corrosion resistance.- The method of test shall be as prescribed in American Society for Testing and Material (ASTM), ASTM B117. The test sample should be air dried for 24 hours prior to testing, to permit sufficient time for the surface finish to harden. The sample shall not be penalized for the presence of white corrosion products on corners or edges. The finish shall not be disturbed. The salt spray solution and exposure shall be as specified in 3.5.1.

4.4.4 Function and casualty.- The method of test shall be as prescribed in AMCR 715-505, Volume 8. The four hundred (400) test sample cases will be assembled as Cartridge, 20MM, Target Practice, Tracer, M206E2, conforming to Dwg. D10542655.* The gun barrel(s) will be at ambient temperature at the

* The test cartridges will be fired in burst of 25 rounds in the 20MM, M139 Gun at the minimum cycle rate of 850 rounds per minute (RPM).

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beginning of the test and will be cooled to ambient temperature after each 100 rounds. This test will be conducted at a facility designated by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.- Each lot of cartridge cases shall be packed as specified by Dwg. F11731525.

5.2 Marking and labeling.- Marking or labeling of shipping containers shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Place of inspection, if not place of manufacture.

6.1.2 First article sample. (see 4.3 and 6.3)

6.1.3 Provision for shipment of production lot acceptance samples for the firing test.

6.1.4 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.5 Provision for the submission and approval of the manufacturing process.

6.2 Process deviation.- A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.3 First article sample.- In order to minimize inspection costs, the firing tests of 4.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridge cases may be required by the testing facility.

6.4 AQL's.- The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

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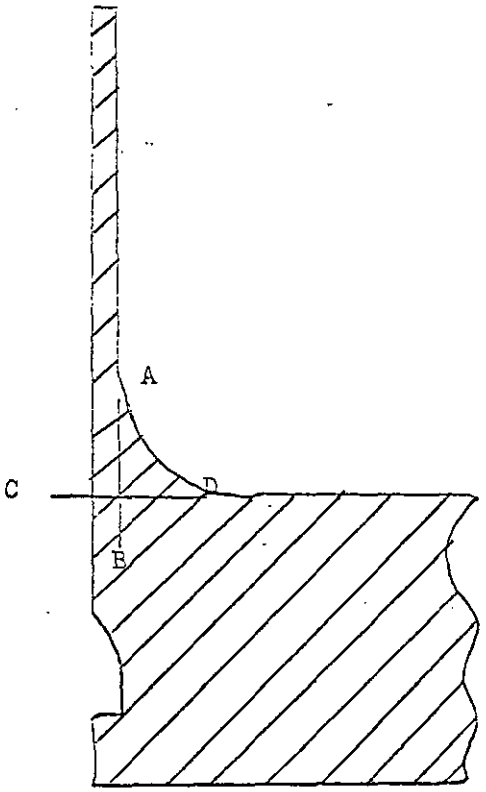
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FIGURE 1



TANGENTS AB AND CD INDICATE LIMITS BEYOND WHICH FOLDS OR COLD SHUTS MUST NOT EXTEND IN ACCEPTABLE CASES.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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